UNIVERSITY EXAMINATIONS: 2013/2014

ORDINARY EXAMINATION FOR THE BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

BIT 2204A JAVA PROGRAMMING

DATE: AUGUST, 2014 **TIME: 2 HOURS**

INSTRUCTIONS: Answer Question ONE and any other TWO

QUESTION ONE

a) Explain any four characteristics that make java language overcome some of the limitations of its predecessors. [8 Marks]

b) Using a diagram, explain the concept of Java platform and its importance. [4 Marks]

c) Write a Java console application that inputs the radius and then computes and displays the area and circumference of the circle. Use console window for inputs and outputs. [4 Marks]

d) Write a Java program that inputs three integer numbers using Inputbox dialogs and computes the sum and average of the numbers. The program should then display the sum and average by use of a message box dialog. Your program should confirm from the user on whether to continue or not. If the user clicks "Yes" then the program computes another set of numbers, and "No" the program exits. [6 Marks]

e) Create a Java application that defines a class named Person. It should have instance variables to record name, age and salary and which are of types String, integer and floating point respectively. Create a Person object and assign and then display the instance variable values using the console window. [5 Marks]

f) Write an applet program that displays the words "Long Live Kenya!!!" at the center of the applet. The words should be displayed using red colour in a yellow background. [3 Marks]

QUESTION TWO

a) Describe the use of the ***this*** keyword in Java. [2 Marks]

b) Among the following statements, identify the ones that will *not* cause a compile-time error. [4 Marks]

i) `int[] a = int[10];`
ii) `int[] a = new int[10];`
iii) `int[10] a = new int[10];`
iv) `int[] a;`
v) `int a = {1, 2, 3};`
vi) `int[][] a = {{1, 2, 3}, {1, 2, 3}};`
vii) `int[][] a = new int[10];`
viii) `int[] a = {1, 2, 3}; int[] b = a;`

c) Consider the following Java program

```
public class HIthere {
     public static void greeting(int x)
     {
          if (x < 10)
          {
               System.out.print("Hello,");
          }
          else
          {
               System.out.print("Bonjour,"); /* French
               for "Hello"*/
          }
          x = x - 5;
     }
     public static void main(String[] args)
     {
          int x = Integer.parseInt(args[0]);
          greeting(x);
          if (x < 3)
          {
               System.out.println(" world."); // Anglais
               //pour "monde"
          }
          else
          {
               System.out.println(" monde.");
          }
     }
}
```

What will be printed by this program in the following cases?

i) java HIthere 10

ii) java HIthere 7 [4 Marks]

d) Write a Java application program to be used to calculate the area and perimeter of the rectangle. Create a Rectangle class with private instance variables length and width and methods to calculate and return the perimeter and area of the rectangle respectively. It also has modifier (set) and accessor (get) methods for inputting and outputting the length and the width. The modifier methods should verify that the length and width are each a floating point number greater that 0.0 and less than 20.0. Define another class that has a main method and used to instantiate the rectangle object and then send relevant messages to compute for area and perimeter of the rectangle. Have both inputs and outputs through console window. [10 Marks]

QUESTION THREE

a) Define the term **Interface** in relation to Java programming. [2 Marks]

b) Explain briefly the significance of using Interface when programming. [3 Marks]

c) Explain two ways in which an interface differs from an abstract class. [3 Marks]

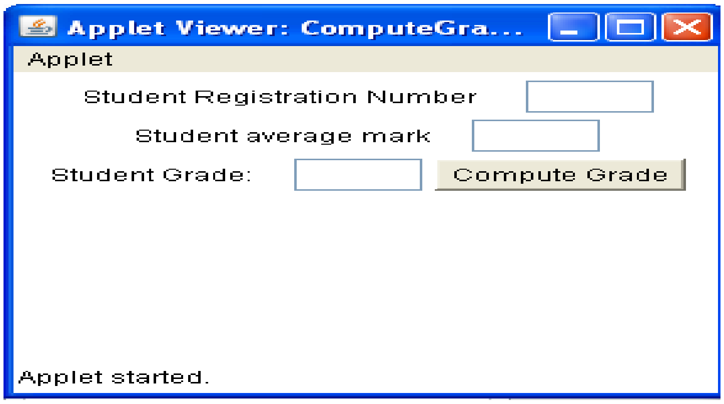
d) Write a Java program to determine the area and perimeter of a circle and rectangle respectively. Your program should consist of an interface and classes that implements the interface and used to compute for the area and perimeter of the circle and rectangle. All classes uses constructors to initialize the data members and have methods area () and perimeter (), which are used to compute for the area and perimeter of the respective shapes. Include a main class to demonstrate the working of the program. [12 Marks]

QUESTION FOUR

a) Explain any four benefits offered by Applets, when developing Internet based applications. [4 Marks]

b) Briefly describe the diffrence between how an applet and frame would implement the same GUI components to render the same look in an iterface. [4 Marks]

c) Write a Java Applet that displays the GUI that contains three text fields, three labels and a button as depicted below. Your applet should allow the student to enter the registration number and average mark in the respective text fields and then grades the student as “Pass” if the average mark is greater than or equal to 50 and “Fail” otherwise after a click of the button. Use the text field labeled “Student Grade” to display the grade. Use the FlowLayout to arrange the GUI widgets. [12 Marks]



QUESTION FIVE

a) Explain the role played by each of the following :

i) Event source

ii) Event object

iii) Event listener [6 Marks]

b) Write a program that adds or subtracts numbers entered by the user, depending on the button clicked. The program should have an interface similar to the one shown below. [14 Marks]